



SCARAB: User Manual

For submission to SETU

Author: Stuart Rossiter

Student Number: C00284845

Course: BSc (Hons.) Software Development

Supervisor: Joseph Kehoe

Requirements

2 Available USB Ports

Arduino IDE

Installation Instructions

PC PROGRAM

Windows:

Download the SCARAB_Setup_Win.exe from GitHub.

Run the installer.

Linux:

Download the SCARAB_Linux.zip file from GitHub.

Extract the zip file, and run SCARAB

SCARAB DEVICE

Download the Arduino IDE from the Arduino website.

Download the SCARAB_SKETCH.zip file from GitHub.

Extract it, and open the SCARAB.ino file with the Arduino IDE.

Plug in your Arduino Mega 2560 you will be using for the SCARAB.

Compile and upload the sketch to the Arduino.

How to Use

Insert the Arduino Mega into the SCARAB Main PCB.

Insert the USB-C Power Cable and USB-A Serial cable into the SCARAB and Arduino respectively.

Insert a module into the SCARAB, with the module facing away from the SCARAB's Main PCB.

Insert a cartridge into the SCARAB's module.

Run the SCARAB PC Program.

If the SCARAB GUI Program is running for the first time, it will not auto-identify the SCARAB. This needs to be enabled in "Options".

Click “Re-Identify SCARAB”. The SCARAB will be identified, along with the Module and Cartridge. If this is a new module, the EEPROM can be written to in “Options”.

Navigate to the “Check Health” menu.

Click “Check Health”.